Task 1: Locality of Different Programs

Configuration

Processors in SMP = 1

Cache coherence protocol = MESI

Scheme for bus arbitration = Random

Word wide (bits) = 16

Words by block = 16 (block size = 32 bytes)

Blocks in main memory = 8192 (main memory size = 256 KB)

Blocks in cache = 128 (cache size = 4 KB)

Mapping = Fully Associative

Replacement policy = LRU

Results

Trace File	Hit Rate	Miss Rate
Hydro	85.190%	14.810%
Nasa7	84.636%	15.364%
Cexp	99.260%	0.740%
Mdljd	85.220%	14.780%
Ear	89.902%	10.098%
Comp	82.092%	17.908%
Wave	82.521%	17.479%
Swm	80.370%	19.630%
UComp	82.730%	17.270%

Questions

1) **Do all the programs have the same locality grade (i.e., local or global)? Which is the program with the best locality? And which does it have the worst? Do you think that the design of memory systems that exploit the locality of certain kind of programs (which will be the most common in a system) can increase the system performance? Why?**

   Although there are some programs with similar locality, there are no programs that have the same locality grade. Cexp (99.26%) demonstrates the best locality with Swm (80.37%) having the worst locality.

   Yes, the design of memory systems that exploit locality will have an increase in performance. The principle of locality reduces the average memory access time by allowing quicker access to data that will be accessed again, or near data that is accessed. Therefore, the overall hit rate will improve, and the miss rate will decrease, leading to better system performance.

2) **During the development of the experiments, you can observe graphically how, in general, the miss rate decreases as the execution of the program goes forward. Why? Which is the reason?**
   At the beginning of each program, compulsory misses cause a spike in miss rates and low hit rates. However, the miss rate decreases near the end of all the experiments because of the principle of locality and the LRU replacement policy. Temporal and spatial locality prefetches cache and accesses contiguous data, reducing the time spent on retrieving data. LRU ensures that only the frequently used blocks remain in the cache.

Task 2: Influence of Cache Size

Configuration

Processors in SMP = 1

Cache coherence protocol = MESI

Scheme for bus arbitration = Random

Word wide (bits) = 16

Words by block = 16 (block size = 32 bytes)

Blocks in main memory = 8192 (main memory size = 256 KB)

Mapping = Fully Associative

Replacement policy = LRU

Results

Trace File: Hydro

Cache Size	Hit rate	Miss rate
1	33.803%	66.197%
2	57.640%	42.360%
4	67.889%	32.111%
8	71.932%	28.068%
16	74.988%	25.012%
32	81.805%	18.195%
64	84.532%	15.468%
128	85.190%	14.810%
256	85.614%	14.386%
512	85.614%	14.386%

Trace file: Nasa7

Cache Size	Hit rate	Miss rate
1	33.423%	66.577%
2	58.976%	41.024%
4	70.350%	29.650%
8	74.178%	25.822%
16	76.765%	23.235%
32	82.965%	17.035%
64	83.935%	16.065%
128	84.636%	15.364%
256	85.013%	14.987%
512	85.013%	14.987%

Trace file: Cexp

Cache Size	Hit rate	Miss rate
1	47.080%	52.920%
2	56.100%	43.900%
4	56.855%	43.145%
8	59.935%	40.065%
16	98.460%	1.540%
32	99.230%	0.770%

64	99.255%	0.745%
128	99.260%	0.740%
256	99.260%	0.740%
512	99.260%	0.740%

Trace file: Mdljd

Cache Size	Hit rate	Miss rate
1	36.945%	63.055%
2	57.435%	42.565%
4	70.180%	29.820%
8	78.425%	21.575%
16	80.285%	19.715%
32	82.115%	17.885%
64	83.960%	16.040%
128	85.220%	14.780%
256	86.765%	13.235%
512	88.775%	11.225%

Trace file: Ear

Cache Size	Hit rate	Miss rate
1	40.053%	59.947%
2	57.856%	42.144%
4	62.378%	37.622%
8	72.231%	27.769%
16	81.782%	18.218%
32	83.007%	16.993%
64	84.326%	15.674%
128	89.902%	10.098%
256	93.293%	6.707%
512	93.293%	6.707%

Trace file: Comp

Cache Size	Hit rate	Miss rate
1	32.311%	65.689%
2	62.203%	37.797%
4	74.604%	25.396%
8	76.743%	23.257%
16	78.645%	21.355%

32	81.339%	18.661%
64	81.815%	18.185%
128	82.092%	17.908%
256	82.092%	17.908%
512	82.726%	17.274%

Trace file: Wave

Cache Size	Hit rate	Miss rate
1	33.645%	66.355%
2	57.864%	42.136%
4	68.077%	31.923%
8	70.907%	29.093%
16	73.125%	26.875%
32	78.436%	21.564%
64	81.646%	18.354%
128	82.521%	17.479%
256	83.192%	16.808%
512	84.680%	15.320%

Trace file: Swm

Cache Size	Hit rate	Miss rate
1	33.130%	66.870%
2	58.620%	41.380%
4	69.165%	30.835%
8	73.915%	26.085%
16	76.125%	23.875%
32	78.015%	21.985%
64	79.185%	20.815%
128	80.370%	19.630%
256	81.715%	18.285%
512	83.940%	16.060%

Trace file: UComp

Cache Size	Hit rate	Miss rate
1	33.955%	66.045%
2	61.764%	38.236%
4	74.119%	24.881%
8	77.534%	22.466%

16	79.510%	20.490%
32	82.108%	17.892%
64	82.473%	17.527%
128	82.730%	17.270%
256	82.730%	17.270%
512	83.315%	16.685%

Questions

1) **Does the miss rate increase or decrease as the cache size increases? Why? Does this increment or decrement happen for all the benchmarks or does it depend on the different locality grades? What does it happen with the capacity and conflict (collision) misses when you enlarge the cache? Are there conflict misses in these experiments? Why?**

   The miss rate decreases as the cache size increases because larger caches can hold more data blocks, reducing the frequency of evictions.

   The decrement happens for all the benchmarks. A larger cache size significantly reduces capacity misses. Since a larger cache can hold more data, caches don't have to be discarded as frequently. A large cache size would also less frequently evict data to make space for new data.

   No, there cannot be any conflict misses in these experiments because the mapping is fully associative. In fully associative caches, there is no fixed mapping and any location is available, so there is no competition for a specific cache location. The only misses are compulsory and capacity.

2) **In these experiments, it may be observed that for great cache sizes, the miss rate is stabilized. Why? We can also see great differences of miss rate for a concrete increment of cache size. What do these great differences indicate? Do these great differences of miss rate appear at the same point for all the programs? Why?**

   The miss rate levels out because as the cache fills up, capacity misses decrease, and the remaining misses will be compulsory misses. Compulsory misses aren't affected by cache size, so increasing cache size will not reduce the miss rate significantly.

   The great differences in miss rate indicate that the cache is large enough to hold the program's working set. No, the great differences in miss rate did not appear at the same time for all programs, and not all programs demonstrated a huge reduction in miss rates. Some programs, like Nasa7, had a steady reduction in miss rate compared to Cexp. This indicates that all the programs have different sizes so a particular cache size that may fit another program may not be sufficient for another program.

3) **In conclusion, does the increase of cache size improve the system performance?**

   Overall, increasing cache size improves the system performance as AMAT improves, due to the reduction in miss rate.

Task 3: Influence of Block Size

Configuration

Processors in SMP = 1

Cache coherence protocol = MESI

Scheme for bus arbitration = Random

Word wide (bits) = 16

Main memory size = 256 KB (the number of blocks in main memory will vary)

Cache size = 4 KB (the number of blocks in cache will vary)

Mapping = Fully Associative

Replacement policy = LRU

Results

Trace File: Hydro

Words By Block	Hit rate	Miss rate
4	56.888%	43.112%
8	75.693%	24.307%
16	85.190%	14.810%
32	90.597%	9.403%
64	93.042%	6.958%
128	94.311%	5.689%
256	95.393%	4.607%
512	93.277%	6.723%
1024	89.375%	10.625%

Trace file: Nasa7

Words By Block	Hit rate	Miss rate
4	55.040%	44.960%
8	74.609%	25.391%
16	84.636%	15.364%
32	90.243%	9.757%
64	92.938%	7.062%
128	94.501%	5.499%
256	96.119%	3.881%
512	94.016%	5.984%
1024	89.919%	10.081%

Trace file: Cexp

Words By Block	Hit rate	Miss rate
4	97.565%	2.435%
8	98.695%	1.305%
16	99.260%	0.740%
32	99.545%	0.455%
64	99.705%	0.295%
128	99.785%	0.215%
256	99.850%	0.150%
512	99.815%	0.185%
1024	98.220%	1.780%

Trace file: Mdljd

Words By Block	Hit rate	Miss rate
4	62.995%	37.005%
8	76.715%	23.285%
16	85.220%	14.780%
32	90.925%	9.075%
64	93.995%	6.005%
128	95.405%	4.595%
256	95.820%	4.180%
512	94.935%	5.065%
1024	89.690%	10.310%

Trace file: Ear

Words By Block	Hit rate	Miss rate
4	79.164%	20.836%
8	86.417%	13.583%
16	89.902%	10.098%
32	92.351%	7.649%
64	94.876%	5.124%
128	95.893%	4.107%
256	96.571%	3.429%
512	93.726%	6.274%
1024	86.078%	13.922%

Trace file: Comp

Words By Block	Hit rate	Miss rate
4	67.552%	32.448%
8	76.902%	23.098%
16	82.092%	17.908%
32	90.095%	9.905%
64	94.097%	5.903%
128	96.434%	3.566%
256	96.712%	3.288%
512	95.959%	4.041%
1024	93.067%	6.933%

Trace file: Wave

Words By Block	Hit rate	Miss rate
4	58.564%	41.436%
8	74.117%	25.883%
16	82.521%	17.479%
32	87.394%	12.606%
64	91.567%	8.433%
128	93.376%	6.624%
256	94.310%	5.690%
512	93.551%	6.449%
1024	91.800%	8.200%

Trace file: Swm

Words By Block	Hit rate	Miss rate
4	46.755%	53.245%

8	68.595%	31.405%
16	80.370%	19.630%
32	87.885%	12.115%
64	91.870%	8.130%
128	93.785%	6.215%
256	94.730%	5.270%
512	93.955%	6.045%
1024	88.505%	11.495%

Trace file: UComp

Words By Block	Hit rate	Miss rate
4	67.069%	32.931%
8	77.205%	22.795%
16	82.730%	17.270%
32	90.450%	9.550%
64	94.329%	5.671%
128	96.597%	3.403%
256	96.963%	3.037%
512	96.304%	3.696%
1024	92.682%	7.318%

Questions

1) **Does the miss rate increase or decrease as the block size increases? Why? Does this increment or decrement happen for all the benchmarks or does it depend on the different locality grades? What does it happen with the compulsory misses when you enlarge the block size? What is the pollution point? Does it appear in these experiments?**

   The miss rate decreases from 4 to 256 words and then increases for 512 and 1024 words across all benchmarks. At the start of the programs, there are compulsory misses and as the cache filles up, the miss rate decreases. However, main memory size (256 KB) and cache size (4 KB) remain constant. As the cache fills up, there will be less space and more capacity conflicts, thus, the increase in miss rate.

   Compulsory misses decrease with larger block sizes because fewer blocks are needed to cover the same memory range.

Yes, there is cache pollution in all the experiments because the miss rate starts increasing from the 512 point. This is an indication that large blocks are starting to pollute the cache with unused data, displacing other useful blocks.

2) **In conclusion, does the increase of block size improve the system performance?**

The increase in block size improves the system performance for a while, but harms performance after a while due to cache pollution.

Task 4: Influence of the Block Size for Different Cache Sizes

Configuration

Processors in SMP = 1

Cache coherence protocol = MESI

Scheme for bus arbitration = Random

Word wide (bits) = 32

Main memory size = 1024 KB (the number of blocks in main memory will vary)

Mapping = Fully Associative

Replacement policy = LRU

Results

Trace file: Ear

Words By Block, Cache size	Hit rate	Miss rate
8, 4	76.112%	23.888%
8, 8	86.417%	13.583%
8, 16	88.338%	11.662%
8, 32	88.357%	11.643%
16, 4	84.326%	15.674%
16, 8	89.902%	10.098%
16, 16	93.293%	6.707%
16, 32	93.293%	6.707%
32, 4	90.298%	9.702%

32, 8	92.351%	7.649%
32, 16	96.138%	3.862%
32, 32	96.157%	3.843%
64, 4	93.463%	6.537%
64, 8	94.876%	5.124%
64, 16	97.607%	2.393%
64, 32	97.664%	2.336%
128, 4	94.857%	5.143%
128, 8	95.893%	4.107%
128, 16	97.739%	2.261%
128, 32	98.531%	1.469%
256, 4	90.995%	9.005%
256, 8	96.571%	3.429%
256, 16	97.532%	2.468%
256, 32	98.945%	1.055%
512, 4	85.625%	14.375%
512, 8	93.726%	6.274%
512, 16	97.268%	2.732%
512, 32	98.417%	1.583%
1024, 4	61.907%	38.093%
1024, 8	86.078%	13.992%
1024, 16	94.367%	5.633%
1024, 32	97.626%	2.374%

Questions

1) **Does the miss rate increase or decrease as the block size increases? Why? What does it happen with the compulsory misses when you enlarge the block size? Does the pollution point appear in these experiments? Does the influence of the pollution point increase or decrease as the cache size increases? Why?**

The miss rate generally decreases as block size increases for each cache size, but pollution eventually occurs at different points depending on cache size. For smaller caches (4 KB), pollution appears at 256 words per block. For larger caches (32 KB), pollution appears at 512 words per block.

Compulsory misses decrease with larger block sizes because each block fetch retrieves more data, reducing the number of first-time accesses needed.

Cache pollution appears in this experiment. The influence of cache pollution increases at 256 and 512 words per block as the cache size increases because the cache reaches max capacity as more data is fetched. The increase in block size improves performance for a while until it reaches a certain point where the cache doesn’t benefit from block size.

Task 5: Influence of the Mapping for Different Cache Sizes

Configuration

Processors in SMP = 1

Cache coherence protocol = MESI

Scheme for bus arbitration = Randon

Word wide (bits) = 32

Words by block = 64 (block size = 256 bytes)

Blocks in main memory = 4096 (main memory size = 1024 KB)

Replacement policy = LRU

Results

Trace file: Ear

Miss rates

	Direct	Two-way	Four-way	Eight-way	Fully associative
4 KB	21.948%	17.257%	18.990%	N/A	18.990%
8 KB	15.109%	10.230%	9.231%	8.327%	8.327%
16 KB	12.340%	8.308%	6.631%	6.594%	6.537%
32 KB	7.027%	6.782%	4.823%	5.256%	5.124%

Questions

1) **Does the miss rate increase or decrease as the associativity increases? Why? What does it happen with the conflict misses when you enlarge the associativity grade? Does the influence of the associativity grade increase or decrease as the cache size increases? Why?**

   The miss rate decreases as the associativity increases because there are more available cache locations for data, reducing conflict misses. Hit rate also improves as the cache can retain and will have more options to store frequently accessed data while replacing the ones less used.

As the associativity increases, the conflict misses decrease. The benefit of associativity decreases as cache size increases. Higher associativity provides more locations for each block, while larger caches can hold more data. These qualities reduce conflict misses as frequently accessed data will likely remain in the cache.

2) **In conclusion, does the increase of associativity improve the system performance? If the answer is yes, in general, which is the step with more benefits: from direct to 2-way, from 2-way to 4-way, from 4-way to 8-way, or from 8-way to fully associative?**

   Yes, the increase in associativity improves system performance. Direct to 2-way showed the largest disparity in improvement, while 8-way to fully associative showed the smallest disparity in improvement.

Task 6: Influence of the Replacement Policy

Configuration

Processors in SMP = 1

Cache coherence protocol = MESI.

Scheme for bus arbitration = Random.

Word wide (bits) = 16

Words by block = 16 (block size = 32 bytes)

Blocks in main memory = 8192 (main memory size = 256 KB)

Blocks in cache = 128 (cache size = 4 KB)

Mapping = 8-way set associative (cache sets = 16)

Results

Miss rates

	Hydro	Nasa7	Cexp	Mdljd	Ear	Comp	Wave	Swm	UComp
Random	15.891%	16.442%	0.855%	15.695%	10.701%	18.542%	19.025%	20.320%	18.149%

LRU	14.763%	15.310%	0.740%	14.790%	9.740%	17.908%	17.362%	19.690%	17.307%
LFU	15.139%	15.849%	0.740%	14.930%	10.004%	17.789%	17.479%	19.860%	17.161%
FIFO	15.139%	15.526%	0.740%	15.095%	10.682%	18.067%	17.741%	20.010%	17.453%

Questions

1) **In general, which is the replacement policy with the best miss rate? And which does it have the worst? Do the benefits of LFU and FIFO policies happen for all the benchmarks or do they depend on the different locality grades?**

   LRU has the best miss rate and random has the worst miss rate.

   LFU and FIFO depend on different locality grades. In Nasa7, FIFO showed better miss rate while LFU had better miss rate for the rest of the programs.

2) **For a direct-mapped cache, would you expect the results for the different replacement policies to be different? Why or why not?**

   For a direct-mapped cache, the replacement policy has no effect on the results because there is no choice in replacement. In direct mapping, each memory block maps to exactly one cache location. When a conflict occurs, the block must be evicted.

3) **In conclusion, does the use of a concrete replacement policy improve the system performance? If the answer is yes, in general, which is the step with more benefits: from Random to LRU, from Random to LFU, or from Random to FIFO? Why (consider the cost/performance aspect)?**

   Yes, the use of a replacement policy improves system performance as it uses the principle of locality to maximize hit rate. Random to FIFO has more benefits. Random is the cheapest replacement policy because it requires the least amount of hardware and computational overhead. LRU and LFU are the most expensive because they need to track and update cache. FIFO has the best performance to cost ratio as it's not as expensive as LRU and LFU but also doesn't have the worst performance compared to random. FIFO and LRU show similar miss rates with small differences. For example, in Cexp and Hydro, FIFO and LRU are the same. The largest difference is 0.678 in Ear.

Random to LRU has the best performance, but most expensive. Random to LFU also has similar performance to LRU, but it's expensive. Random to FIFO is not only as expensive as LRU and LFU, but it also has good performance.